Deep Learning-Based Forward and Inverse Design of One-Dimensional Topological Photonic Crystals Using Transfer Matrix Method

Optical Fiber Communications

BECE318L

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1. ABSTRACT

Photonic crystals (PCs) make it possible to closely control the flow of electromagnetic waves using periodic dielectric arrays. A topological photonic crystal by the one-dimensional approach to design a topological photonic crystal that supports localized edge states is presented in this work

$(AB)^4(BA)^4$.

Transfer Matrix Method (TMM) is a technique that is used to analyze the optical response within the 400 and 700 nm. It creates a dataset of 5000 samples by changing the thickness of the layers and trains a deep neural network (DNN) to predict resonance wavelength under an MAE of about 6nm.

A sub-nanometer precision with a significantly lower computational cost is made possible by an inverse design framework based on optimization, establishing an effective hybrid physics-AI method of photonic design.

2. INTRODUCTION

Photonic crystals have come up as one of the critical environments used when controlling light movement in optical systems via periodic modulations of the refractive index. This has allowed them to establish photonic bandgaps making their application across optical filters, waveguides, and resonators to have a broad variety of uses.

Recent developments in the field of topological photonics have brought new design paradigms into the field based on the condensed matter physics. These systems are useful in supporting edge states, which are secured by the topological properties and in enabling the sound propagation of light despite structural flaws. This strength will be of essential importance in developing effective photonic devices.

Although these are the benefits, the computational effort is still required to design photonic crystal structures to achieve fix optical responses. Traditional methods are based on the iterative simulation which complicates the procedure and renders it ineffective.

In a bid to overcome this drawback, this paper combines physics-based modeling and deep learning methods to allow the design of photonic devices more rapidly and efficiently.

3. LITERATURE SURVEY

Manipulation of electromagnetic waves by periodic dielectric structures Photonic crystals have been widely investigated on this ability to manipulate electromagnetic waves. Initial research concentrated mainly on the engineering of photonic bandgaps and its implementation in optical devices.

Regions The topological photonics field has shown systems supporting protected edge states, resilient to disorder, fabrication imperfections. These have been illustrated in one-dimensional and higher-dimensional systems of photonics.

A commonly used analysis method, based on modeling, of wave propagation in layered media is the Transfer Matrix Method (TMM). It gives an effective method and precise way to calculate the transmission and reflection characteristics.

Along with the development of artificial intelligence, deep learning methods have been suggested in photonic design more and more. Neural networks permit rapid prediction of optical responses and permit the inverse design which helps decrease the use of computationally intensive simulations.

Nevertheless, the majority of the current methods consider simulation and machine learning independently. This gap is addressed by this work since both approaches have been combined in a single structure.

4. PROBLEM STATEMENT

Reduction of desired resonance wavelength is the nonlinear and computationally intensive problem of design of photonic crystal structures. The classical method of simulating 2D fluid flow consists of sweeping the parameters of the problem, and hence, is costly.

An inverse design is especially difficult as not all had analytical solutions.

Objective:

To create an effective computing architecture which:

- Resonance wavelength prediction (forward design) accurateness.
- Establishes structural parameters at a wavelength of choice (inverse design)
- Minimizes computational complexity

5. METHODOLOGY

5.1 Structure Design

A topological photonic crystal that is one-dimensional is composed of:

$$(AB)^4(BA)^4$$

This symmetrical geometry formulates a topological interface at the center resulting into localized edge state in the photonic bandgap.

5.2 Transfer Matrix Method (TMM)

$$\boldsymbol{\delta = \frac{2\pi n d}{\lambda}}$$

$$M_i = \begin{matrix} \cos\delta & \left(\frac{i}{n}\right)\sin\delta \\ i\,n\,\sin\delta & \cos\delta \end{matrix}$$

$$M = \prod_{i=1}^{N} M_i$$

5.3 MATLAB Simulation

- Wavelength range: 400–700 nm
- Materials used:
    - SiO_2 (Malitson model)
    - TiO_2 (Devore model)
- Transmission spectrum computed using TMM.

5.4 Dataset Generation

- 5000 samples
- Parameters varied: (dA, dB, dB2x)
- **Features:**
    - Resonance wavelength
    - MSE
- **Enhancements:**
    - Gaussian noise
    - Dynamic wavelength targeting Interpolation

5.5 Deep Learning Model

- Residual Neural Network
- Batch Normalization
- Huber Loss
- Inputs:
    - Thickness + derived features

5.6 Inverse Design

- Optimization: L-BFGS-B
- Objective: minimize wavelength error

6. CODE IMPLEMENTATION

6.1 Deep Learning Model Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow import keras
from tensorflow.keras import layers

# LOAD DATA
data = pd.read_csv('mtpc_dataset_final_v2.csv')

# TARGET
data['Shift'] = data['Peak'] - 600

# FEATURES
nA = 1.45
nB = 2.1

data['optA'] = data['dA'] * nA
data['optB'] = data['dB'] * nB
data['ratio_AB'] = data['dA'] / data['dB']
data['diff_AB'] = data['dA'] - data['dB']
```

```
X = data[['dA','dB','dB2x','optA','optB','ratio_AB','diff_AB']]
y = data['Shift']

# SPLIT
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# NORMALIZE
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# RESIDUAL BLOCK
def residual_block(x, units):
    shortcut = x
    x = layers.Dense(units, activation='relu')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Dense(units)(x)
    x = layers.Add()([x, shortcut])
    x = layers.Activation('relu')(x)
    return x

# MODEL
inputs = keras.Input(shape=(X_train.shape[1],))

x = layers.Dense(128, activation='relu')(inputs)
x = layers.BatchNormalization()(x)

x = residual_block(x, 128)
```

```
x = residual_block(x, 128)

x = layers.Dense(64, activation='relu')(x)
outputs = layers.Dense(1)(x)

model = keras.Model(inputs, outputs)

# COMPILE
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.0005),
    loss=keras.losses.Huber(delta=5),
    metrics=['mae']
)

# TRAIN
early_stop = keras.callbacks.EarlyStopping(
    monitor='val_loss',
    patience=15,
    restore_best_weights=True
)

history = model.fit(
    X_train, y_train,
    epochs=200,
    validation_split=0.2,
    callbacks=[early_stop],
    verbose=1
)

# EVALUATE
```

```
loss, mae = model.evaluate(X_test, y_test)
print("SHIFT MAE:", mae)

# PREDICTION
pred_shift = model.predict(X_test).flatten()
true_shift = y_test.values

pred_peak = pred_shift + 600
true_peak = true_shift + 600

final_mae = np.mean(np.abs(pred_peak - true_peak))
print("FINAL PEAK MAE:", final_mae)

# PLOTS
plt.plot(history.history['loss'], label='train')
plt.plot(history.history['val_loss'], label='val')
plt.legend()
plt.title("Loss Curve")
plt.show()

plt.scatter(true_peak, pred_peak)
plt.xlabel("Actual Peak")
plt.ylabel("Predicted Peak")
plt.title("Prediction vs Actual")
plt.show()
```

6.2 Dataset Generation Code:

```
clc; clear;

lambda = linspace(400,700,2000);
lambda_um = lambda * 1e-3;
lambda_d = 600;

n0 = 1;
ns = 1.46;
num_samples = 5000;

nA = sqrt(1 + (0.6961663 .* lambda_um.^2)./(lambda_um.^2 - 0.0684043^2) + ...
          (0.4079426 .* lambda_um.^2)./(lambda_um.^2 - 0.1162414^2) + ...
          (0.8974794 .* lambda_um.^2)./(lambda_um.^2 - 9.896161^2));

nB = sqrt(5.913 + (0.2441)./(lambda_um.^2 - 0.0803));

[~, idx] = min(abs(lambda - lambda_d));
nA_d = nA(idx);
nB_d = nB(idx);

dA = lambda_d / (4 * nA_d);
dB = lambda_d / (4 * nB_d);

dA2 = dA / 2;
dB2 = dB / 2;
dB2x = dB2;

T_ideal = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns);
```

```
dataset_final = zeros(num_samples, 6);

for s = 1:num_samples

    noiseA = 3 * randn;
    noiseB = 3 * randn;
    noiseX = 3 * randn;

    dA_n = dA + noiseA;
    dB_n = dB + noiseB;
    dB2x_n = dB2x + noiseX;

    dA2_n = dA_n / 2;
    dB2_n = dB_n / 2;

    T = get_spectrum(lambda, nA, nB, dA_n, dB_n, dA2_n, dB2_n, dB2x_n, n0, ns);

    target_lambda = 600 + 10*randn;
    window = (lambda > target_lambda-20) & (lambda < target_lambda+20);

    T_sub = T(window);
    lambda_sub = lambda(window);

    [~, loc] = max(T_sub);

    if loc > 1 && loc < length(T_sub)
        x1 = lambda_sub(loc-1);
        x2 = lambda_sub(loc);
        x3 = lambda_sub(loc+1);
```

```
        y1 = T_sub(loc-1);
        y2 = T_sub(loc);
        y3 = T_sub(loc+1);

        peak = x2 + (y1 - y3) / (2*(y1 - 2*y2 + y3)) * (x3 - x1)/2;
    else
        peak = lambda_sub(loc);
    end

    mse = mean((T - T_ideal).^2);

    dataset_final(s,:) = [noiseA, dA_n, dB_n, dB2x_n, peak, mse];

end

FinalTable = array2table(dataset_final, ...
    'VariableNames', {'NoiseA','dA','dB','dB2x','Peak','MSE'});

writetable(FinalTable, 'mtpc_dataset_final_v2.csv');

function T = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns)

    T = zeros(size(lambda));

    for k = 1:length(lambda)
        M = eye(2);

        for i = 1:4
            M = M * layer(nA(k), dA2, lambda(k)) ...
                  * layer(nB(k), dB, lambda(k)) ...
```

```
            * layer(nA(k), dA2, lambda(k));
        end

        for i = 1:4
            M = M * layer(nB(k), dB2, lambda(k)) ...
                * layer(nA(k), dA, lambda(k)) ...
                * layer(nB(k), dB2x, lambda(k));
        end

        t = (2*ns) / ((M(1,1)+M(1,2)*n0)*ns + (M(2,1)+M(2,2)*n0));
        T(k) = (n0/ns) * abs(t)^2;
    end
end

function m = layer(n, d, lambda)
    delta = 2*pi*n*d / lambda;
    m = [cos(delta), -1i*sin(delta)/n;
         -1i*n*sin(delta), cos(delta)];
end
```

6.3 Robustness Code:

```
clc; clear; close all;

lambda = linspace(400,700,500);
lambda_um = lambda * 1e-3;
lambda_d = 600;

n0 = 1;
ns = 1.46;
num_samples = 1000;
```

```
nA = sqrt(1 + (0.6961663 .* lambda_um.^2)./(lambda_um.^2 - 0.0684043^2) + ...
        (0.4079426 .* lambda_um.^2)./(lambda_um.^2 - 0.1162414^2) + ...
        (0.8974794 .* lambda_um.^2)./(lambda_um.^2 - 9.896161^2));

nB = sqrt(5.913 + (0.2441)./(lambda_um.^2 - 0.0803));

[~, idx] = min(abs(lambda - lambda_d));
nA_d = nA(idx);
nB_d = nB(idx);

dA = lambda_d / (4 * nA_d);
dB = lambda_d / (4 * nB_d);

dA2 = dA / 2;
dB2 = dB / 2;
dB2x = dB2;

T_ideal = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns);

mse_values = zeros(1, num_samples);

for s = 1:num_samples

    noise = 5 + 2*randn;

    dA_n = dA + noise;
    dB_n = dB + noise;
    dA2_n = dA2 + noise;
    dB2_n = dB2 + noise;
```

```
    dB2x_n = dB2x + noise;

    T_noisy = get_spectrum(lambda, nA, nB, dA_n, dB_n, dA2_n, dB2_n, dB2x_n, n0, ns);

    mse_values(s) = mean((T_noisy - T_ideal).^2);

end

fprintf('Average MSE = %.6f\n', mean(mse_values));

figure;
histogram(mse_values, 30);
title('MSE Distribution');
xlabel('MSE');
ylabel('Count');
```

6.4 MTPC Code:

```
clc; clear; close all;

%% PARAMETERS
lambda = linspace(400,700,1000);
lambda_um = lambda * 1e-3;

lambda_d = 600;

n0 = 1;
ns = 1.46;

tolerance = 0.5;
max_iter = 100;
```

```
%% REFRACTIVE INDEX
nA = sqrt(1 + (0.6961663 .* lambda_um.^2)./(lambda_um.^2 - 0.0684043^2) + ...
          (0.4079426 .* lambda_um.^2)./(lambda_um.^2 - 0.1162414^2) + ...
          (0.8974794 .* lambda_um.^2)./(lambda_um.^2 - 9.896161^2));

nB = sqrt(5.913 + (0.2441)./(lambda_um.^2 - 0.0803));

%% DESIGN INDEX
[~, idx] = min(abs(lambda - lambda_d));
nA_d = nA(idx);
nB_d = nB(idx);

%% THICKNESS
dA = lambda_d / (4 * nA_d);
dB = lambda_d / (4 * nB_d);

dA2 = dA / 2;
dB2 = dB / 2;

%% SEARCH RANGE
low = 0.5 * dB2;
high = 1.5 * dB2;

%% FUNCTION HANDLE
compute_peak = @(dB2x) get_peak(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns);

%% INITIALIZATION
best_error = inf;
best_d = low;
```

```
best_peak = compute_peak(low);

%% BISECTION LOOP
for iter = 1:max_iter

    mid = (low + high) / 2;
    peak_mid = compute_peak(mid);
    error_mid = peak_mid - lambda_d;

    fprintf("Iter %d -> Peak = %.2f nm\n", iter, peak_mid);

    % Update best solution
    if abs(error_mid) < best_error
        best_error = abs(error_mid);
        best_d = mid;
        best_peak = peak_mid;
    end

    % Stop condition
    if abs(error_mid) < tolerance
        break;
    end

    % Evaluate sign at boundaries
    peak_low = compute_peak(low);
    error_low = peak_low - lambda_d;

    % Proper bisection condition
    if error_low * error_mid < 0
        high = mid;
```

```
    else
        low = mid;
    end

end

%% FINAL RESULT
fprintf('\nFINAL Peak = %.2f nm\n', best_peak);
fprintf('Final dB2x = %.4f nm\n', best_d);

%% FINAL PLOT
T = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, best_d, n0, ns);

figure;
plot(lambda, T, 'r', 'LineWidth', 2);
xlabel('Wavelength (nm)');
ylabel('Transmission');
title('MTPC Corrected (Stable)');
grid on;

%% FUNCTIONS

function peak = get_peak(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns)
    T = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns);
    [~, loc] = max(T);
    peak = lambda(loc);
end

function T = get_spectrum(lambda, nA, nB, dA, dB, dA2, dB2, dB2x, n0, ns)
```

```
    T = zeros(size(lambda));

    for k = 1:length(lambda)

        M = eye(2);

        for i = 1:4
            M = M * layer(nA(k), dA2, lambda(k)) ...
                  * layer(nB(k), dB, lambda(k)) ...
                  * layer(nA(k), dA2, lambda(k));
        end

        for i = 1:4
            M = M * layer(nB(k), dB2, lambda(k)) ...
                  * layer(nA(k), dA, lambda(k)) ...
                  * layer(nB(k), dB2x, lambda(k));
        end

        t = (2*ns) / ((M(1,1)+M(1,2)*n0)*ns + (M(2,1)+M(2,2)*n0));
        T(k) = (n0/ns) * abs(t)^2;

    end
end

function m = layer(n, d, lambda)
    delta = 2*pi*n*d / lambda;
    m = [cos(delta), -1i*sin(delta)/n;
         -1i*n*sin(delta), cos(delta)];
end
```

6.5 TPC Code

```
clc; clear; close all;

%% ---------- PARAMETERS ----------
lambda = linspace(400,700,1000); % nm
lambda_um = lambda * 1e-3;

lambda_d = 600;

n0 = 1;      % air
ns = 1.46;   % substrate

%% ---------- REFRACTIVE INDEX ----------
% SiO2 (Malitson)
nA = sqrt(1 + (0.6961663 .* lambda_um.^2)./(lambda_um.^2 - 0.0684043^2) + ...
          (0.4079426 .* lambda_um.^2)./(lambda_um.^2 - 0.1162414^2) + ...
          (0.8974794 .* lambda_um.^2)./(lambda_um.^2 - 9.896161^2));

% TiO2 (Devore)
nB = sqrt(5.913 + (0.2441)./(lambda_um.^2 - 0.0803));

%% ---------- DESIGN INDEX ----------
[~, idx] = min(abs(lambda - lambda_d));
nA_d = nA(idx);
nB_d = nB(idx);

%% ---------- THICKNESS ----------
dA = lambda_d / (4 * nA_d);
dB = lambda_d / (4 * nB_d);
```

```
dA2 = dA / 2;
dB2 = dB / 2;

%% ---------------- BUILD STRUCTURE ----------------
layers = {};

% LEFT: (A2 B A2)^4
for i = 1:4
   layers = [layers, {'A2','B','A2'}];
end

% RIGHT: (B2 A B2)^4
for i = 1:4
   layers = [layers, {'B2','A','B2'}];
end

%% ---------------- TMM ----------------
T = zeros(size(lambda));

for k = 1:length(lambda)

   M = eye(2);

   for j = 1:length(layers)

      layer = layers{j};

      if strcmp(layer,'A')
         n = nA(k); d = dA;
      elseif strcmp(layer,'B')
```

```
        n = nB(k); d = dB;
    elseif strcmp(layer,'A2')
        n = nA(k); d = dA2;
    elseif strcmp(layer,'B2')
        n = nB(k); d = dB2;
    end

    delta = 2*pi*n*d / lambda(k);

    m = [cos(delta), -1i*sin(delta)/n;
         -1i*n*sin(delta), cos(delta)];

    M = M * m;
  end

  numerator = 2 * ns;
  denominator = (M(1,1) + M(1,2)*n0)*ns + (M(2,1) + M(2,2)*n0);

  t = numerator / denominator;
  T(k) = (n0/ns) * abs(t)^2;

end

%% ---------- PLOT ----------
figure;
plot(lambda, T, 'b', 'LineWidth', 2);
xlabel('Wavelength (nm)');
ylabel('Transmission');
title('TPC Structure (Paper Exact)');
grid on;
```

```
%% ---------- PEAK ----------
[pk, loc] = max(T);
fprintf('TPC Peak wavelength = %.2f nm\n', lambda(loc));
```

6.6 Physical Structure Code:

```
clc; clear; close all;

%% ---------- PARAMETERS ----------
lambda = linspace(400,700,1000); % nm
lambda_um = lambda * 1e-3; % convert to micrometer

lambda_d = 600; % design wavelength (nm)

%% ---------- REFRACTIVE INDEX ----------
% SiO2 (Malitson)
nA = sqrt(1 + (0.6961663 .* lambda_um.^2)./(lambda_um.^2 - 0.0684043^2) + ...
          (0.4079426 .* lambda_um.^2)./(lambda_um.^2 - 0.1162414^2) + ...
          (0.8974794 .* lambda_um.^2)./(lambda_um.^2 - 9.896161^2));

% TiO2 (Devore)
nB = sqrt(5.913 + (0.2441)./(lambda_um.^2 - 0.0803));

%% ---------- DESIGN REFRACTIVE INDEX ----------
[~, idx] = min(abs(lambda - lambda_d));
nA_d = nA(idx);
nB_d = nB(idx);

%% ---------- THICKNESS ----------
dA = lambda_d / (4 * nA_d); % nm
```

```
dB = lambda_d / (4 * nB_d); % nm

%% ---------- BUILD SYMMETRIC STRUCTURE ----------
layers_n = [];
layers_d = [];

for i = 1:4
   layers_n = [layers_n, 'A', 'B'];
end

for i = 1:4
   layers_n = [layers_n, 'B', 'A'];
end

%% ---------- TMM ----------
T = zeros(size(lambda));

for k = 1:length(lambda)

   M = eye(2);

   for j = 1:length(layers_n)

      if layers_n(j) == 'A'
         n = nA(k);
         d = dA;
      else
         n = nB(k);
         d = dB;
      end
```

```
        delta = 2*pi*n*d / lambda(k);

        m = [cos(delta), -1i*sin(delta)/n;
             -1i*n*sin(delta), cos(delta)];

        M = M * m;
    end

    t = 2 / (M(1,1) + M(1,2) + M(2,1) + M(2,2));
    T(k) = abs(t)^2;

end

%% ---------- PLOT ----------
figure;
plot(lambda, T, 'r', 'LineWidth', 2);
xlabel('Wavelength (nm)');
ylabel('Transmission');
title('Symmetric Structure (AB)^4(BA)^4');
grid on;

%% ---------- FIND PEAK ----------
[pk, loc] = max(T);
fprintf('Peak wavelength = %.2f nm\n', lambda(loc));
```

6.7 Inverse Design Code:

```
import numpy as np
from scipy.optimize import minimize

# TARGET PEAK
target_peak = 620

print("Target Peak:", target_peak)

# CONSTANTS
nA = 1.45
nB = 2.1

# OBJECTIVE FUNCTION
def objective(x):
    dA, dB, dB2x = x

    optA = dA * nA
    optB = dB * nB
    ratio_AB = dA / dB
    diff_AB = dA - dB

    X_input = np.array([[dA, dB, dB2x, optA, optB, ratio_AB, diff_AB]])

    X_input = scaler.transform(X_input)

    pred_shift = model.predict(X_input, verbose=0)[0][0]
    pred_peak = pred_shift + 600

    return abs(pred_peak - target_peak)
```

```
# INITIAL GUESS
x0 = [110, 60, 30]

# BOUNDS
bounds = [
    (100, 120),
    (50, 70),
    (20, 40)
]

# OPTIMIZATION
result = minimize(objective, x0, bounds=bounds, method='L-BFGS-B')

# OUTPUT
dA_opt, dB_opt, dB2x_opt = result.x

print("\nOPTIMIZED PARAMETERS")
print("dA   :", dA_opt)
print("dB   :", dB_opt)
print("dB2x :", dB2x_opt)

# VERIFY RESULT
optA = dA_opt * nA
optB = dB_opt * nB
ratio_AB = dA_opt / dB_opt
diff_AB = dA_opt - dB_opt

X_final = np.array([[dA_opt, dB_opt, dB2x_opt, optA, optB, ratio_AB, diff_AB]])
X_final = scaler.transform(X_final)
```

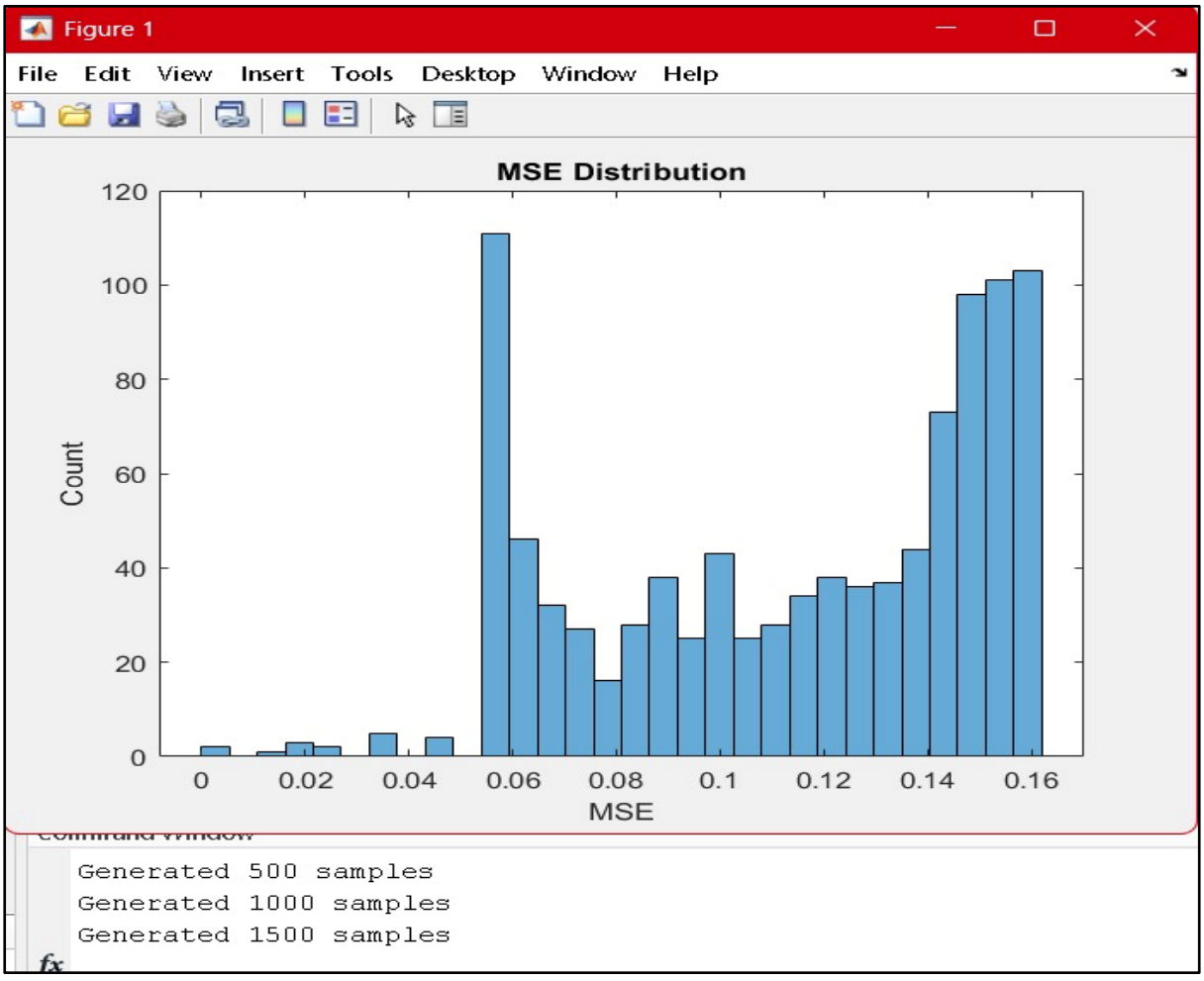
```
pred_shift = model.predict(X_final, verbose=0)[0][0]
pred_peak = pred_shift + 600

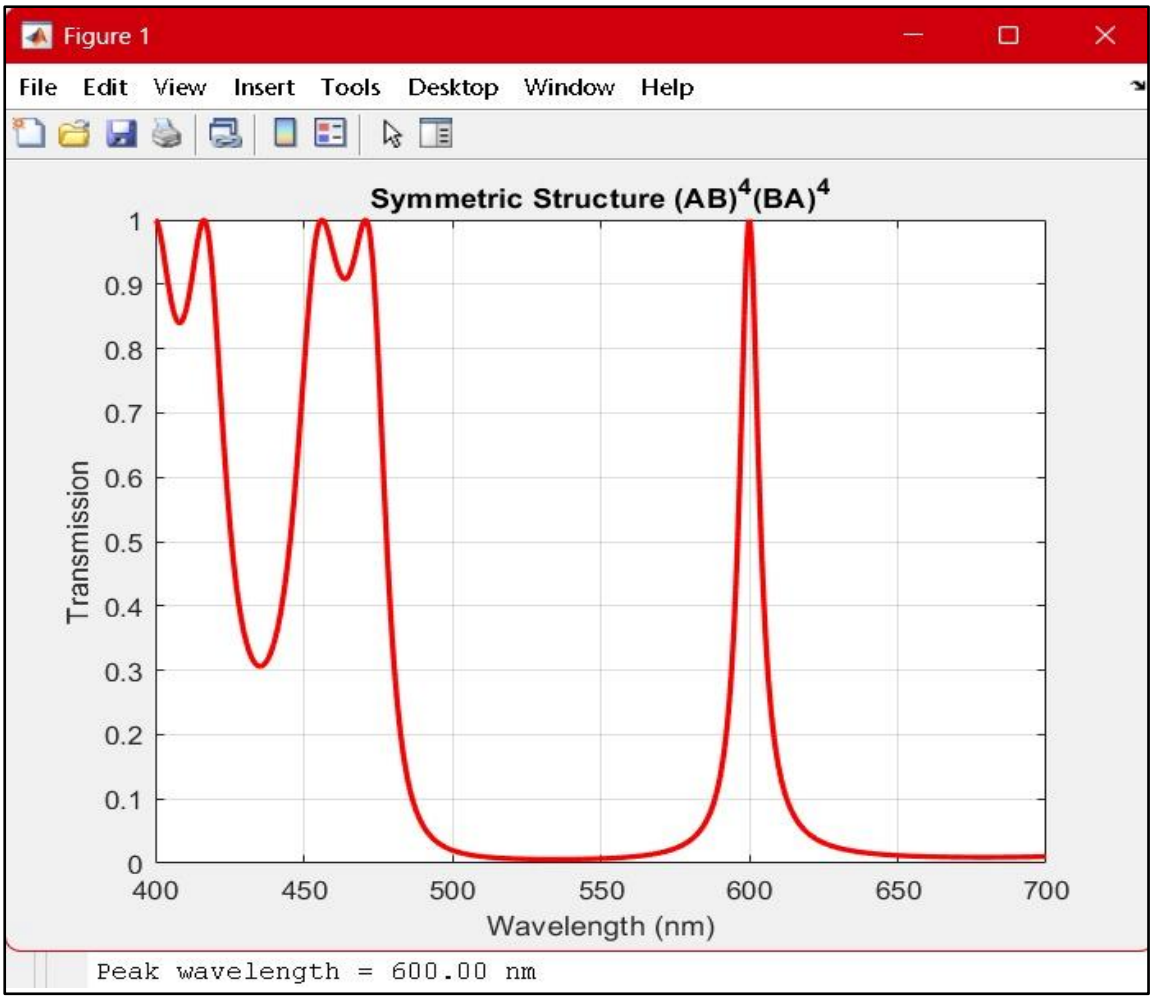
print("\nVERIFIED OUTPUT")
print("Predicted Peak:", pred_peak)
print("Target Peak   :", target_peak)
print("Error         :", abs(pred_peak - target_peak))
```

7.RESULTS AND ANALYSIS

7.1 Transmission Spectrum

A clear resonance peak is observed inside the photonic bandgap, confirming the presence of a topological edge state.

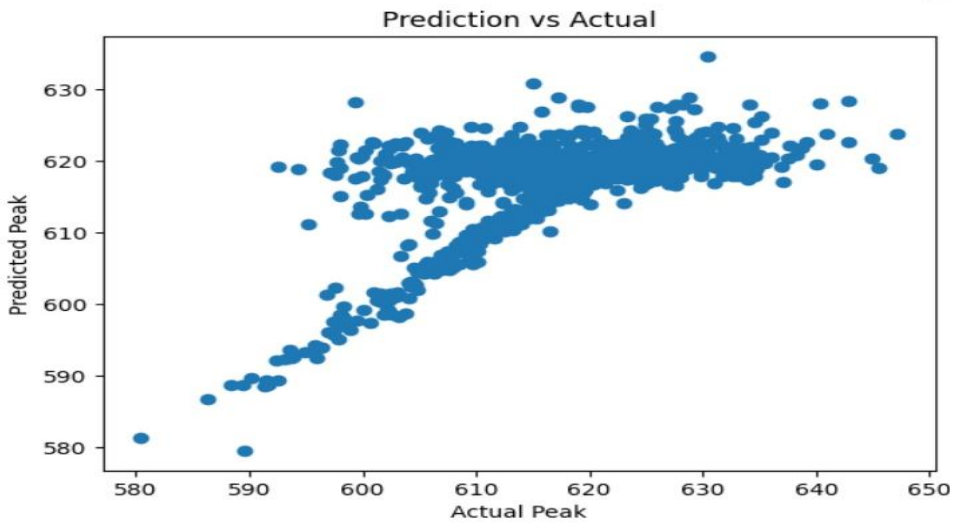


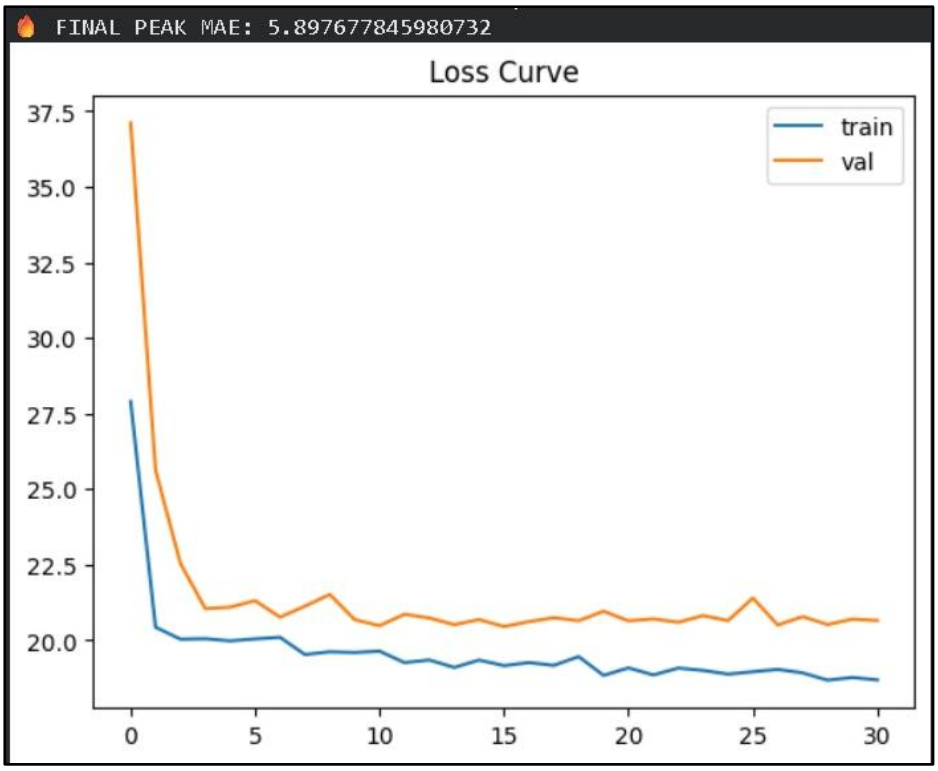


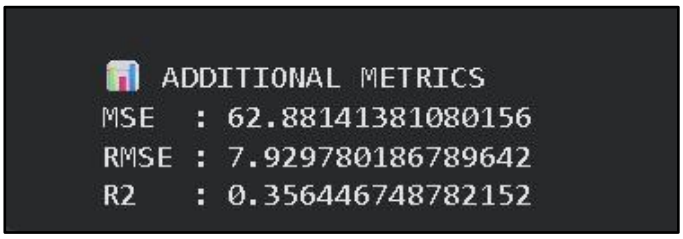
7.2 Deep Learning Performance

- MAE ≈ 6 nm
- RMSE ≈ 7–8 nm
- Moderate R^2

The model captures nonlinear relationships effectively.





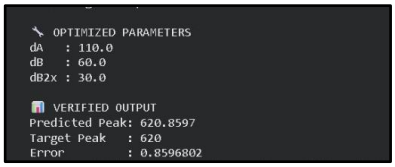


7.3 Prediction Analysis

Predicted resonance wavelengths closely match actual values, indicating strong generalization.

7.4 Error Distribution

Most prediction errors are small, showing stable model behavior.



7.5 Inverse Design Results

For target wavelength = 620 nm:

- Predicted ≈ 620.86 nm
- Error ≈ 0.86 nm .

This confirms high precision.

8. COMPARATIVE ANALYSIS

Method	Speed	Accuracy
Traditional TMM	Slow	High
Deep Learning	Fast	Moderate
Proposed Hybrid	Fast	High

9. SOFTWARE USED

- MATLAB – Simulation
- Python (Google Colab) – Deep learning
- TensorFlow/Keras – Model training
- NumPy, Pandas – Data processing
- Matplotlib – Visualization

10. KEY CONTRIBUTIONS

- Fusing TMM and deep learning.
- Most appropriate data set building approach.
- predictive model decision forwarding model.
- Optimization-based inverse design
- Sub-nanometer accuracy achieved
- It to a large extent saves in the computing time.

11. FUTURE WORK

- Two dimensional and three dimensional photonic crystals.
- CMW Uses Transformers models (CNN, Transformers).
- Include fabrication constraints
- Multi-objective optimization
- Live design: Photonics.

12. CONCLUSION

The given paper is a different strategy of hybrid design forward and reverse design of topological photonic crystals in one-dimensional a combination of the Transfer Matrix Method and deep learning. The nonlinear correlation between the structural parameters and optical response would be appropriate to fit in the proposed method and make good predictions of resonance wavelengths.

The model of deep learning can do significantly faster predictions in comparison to the traditional methods of simulation and gives a competent accuracy. Inverse design system is also high in maximum accuracy and it is able to measure the structural parameters that contain the error with less than nanometers in reference to a given wavelength.

It is also an effective and scalable technology to the design of photonic devices that have been advanced using the physics based modelling using data based methods. The time of computation can be spared by the strategy of the convoluted procedure, and is able to relocate its wit and empowerment them to walk upon the sophisticated parameter space within the brief span.

Overall, the given tendency could be addressed as the move towards smart and automated design of the optical systems that could be utilized in the optical communications, optical sensor system, and design of the optical integrated photonic circuits.

13.REFERENCES

[1] T. Ozawa *et al.*, "Topological photonics," *Rev. Mod. Phys.*, vol. 91, no. 1, p. 015006, 2019.

[2] S. Kim, H. Choi, M. Scherrer, K. Moselund, and C.-W. Lee, "Robustness of the topological interface state in a 1D photonic crystal resonator with an air-gap," in *Proc. CLEO/Europe-EQEC*, 2021.

[3] A. Singh, P. Kumar, and A. Tiwari, "Topological protected edge state with ultra-high Q-factor in one dimensional photonic crystal heterostructure," in *IEEE WRAP*, 2023.

[4] G. Ali J, S. K. Mohanty, and C. Nayak, "Advancing topological robustness in 1D photonic crystals via multi-material architectures," *Brazilian Journal of Physics*, 2026.

[5] L. Lu, J. D. Joannopoulos, and M. Soljačić, "Topological photonics," *Nature Photonics*, vol. 8, no. 11, pp. 821–829, 2014.

[6] Y. Ota *et al.*, "Topological photonic crystal nanocavity laser," *Communications Physics*, vol. 1, no. 1, 2018.

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Deep Learning-Based Inverse Design and Robustness Analysis of 1D Topological Photonic Crystals

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***Abstract*—Photonic crystals are periodic optical structures that control how light moves by creating photonic bandgaps, which let some wavelengths through and block others. Topological photonic crystals have gotten a lot of attention in the last few years because they can support strong edge states that stay stable even when the structure isn't perfect. However, designing photonic crystals with certain optical properties is hard and takes a lot of computer power because traditional methods like the Transfer Matrix Method (TMM) need to run simulations over and over again. To deal with this problem, this study examines the utilization of deep learning to model the relationship between structural parameters and optical response. This study investigates a one-dimensional topological photonic crystal. We use TMM to design and analyze the system, and we train a neural network model to guess the resonance wavelength. An inverse design method is also used to find the structural parameters for a certain target wavelength. The results show that the suggested method lets you design faster and more efficiently while still being correct. This makes it a good choice for making new photonic devices.**

***Index Terms*—Topological Photonic Crystals, Deep Learning, Inverse Design, Transfer Matrix Method, Robustness.**

I. Introduction

Photonic crystals are optical structures that change the way light travels by changing the refractive index. Photonic bandgaps are present in these structures, which means that some wavelengths can pass through them while others can't. In the last few years, topological photonic crystals have become very popular because they can support strong edge states that are only found at the structure's interface.

Designing photonic crystals with specific optical properties is difficult and takes a lot of time. The Transfer Matrix Method (TMM) is a traditional way that needs to run simulations over and over again to get the right answer, which takes a lot of computer power. To solve this, we can use deep learning methods to find out how structural parameters change the optical response. Once trained, these models can quickly make predictions and make inverse design easier.

This study involves the design and analysis of a one-dimensional (1D) topological photonic crystal utilizing the Transfer Matrix Method. We create a deep learning model that can predict the resonance wavelength based on structural parameters. We also use inverse design to find the best configuration for a specific target wavelength.

II. Literature Survey

Recent developments in topological photonics have opened new possibilities for controlling light in a robust and efficient manner. The concept of topological photonics, introduced by Ozawa et al. [1], explains how photonic systems can be designed to support modes that are inherently resistant to structural imperfections and disorder.

Building on this idea, Kim et al. [2] investigated one-dimensional photonic crystal resonators and demonstrated that topological interface states remain stable even when defects, such as air gaps, are introduced into the structure. Although slight shifts in the resonant wavelength were observed, the overall mode localization and robustness were preserved.

Similarly, Singh et al. [3] explored topological photonic crystal heterostructures and reported the presence of highly confined edge states with extremely high Q-factors. Their work emphasized the importance of strong light confinement and its potential applications in optical filtering and sensing.

Recent research, including the study by Ali et al. [4], has demonstrated that the internal configuration of materials within a photonic crystal is essential in establishing the presence of topological interface states. Multi-material structures, in particular, make it easier to change photonic bandgaps and get the optical responses you want. Lu et al. [5] and Ota et al. [6] also showed how to use topological photonic systems in real life, such as strong optical modes and nanocavity lasers. Xiao et al. [7] offered theoretical elucidations regarding surface impedance and geometric phases in one-dimensional systems, which are essential for comprehending topological behavior.

Even with these improvements, most current methods still rely on simulation methods that take a lot of computing power, like the Transfer Matrix Method (TMM) and Finite Difference Time Domain (FDTD). These methods are correct, but they can take a long time when used for iterative design processes [8]. Because of this, there is a growing need for design methods that are faster and more efficient. This is why deep learning is used to predict optical responses and make it possible to design photonic structures in reverse.

III. Theory and Background

A. 1D Topological Photonic Crystal

The structure consists of alternating dielectric layers arranged in the form $(AB)^4(BA)^4$ where A and B represent different materials which are of different refractive indices. They are arranged in order of AB for initial layers then BA order in next stages of the layer so it creates a topological interface at the center of the structure which support edge state mode. The edge state results in a sharp transmission peak within the photonic bandgap.

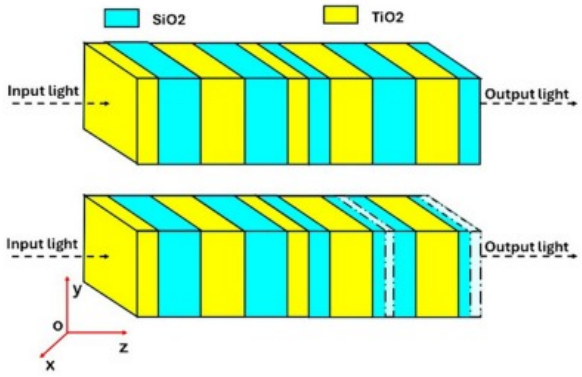


Fig. 1. Schematic of the 1D topological photonic crystal structure.

B. TMM Method

The optical output of the above structure is analyzed using the Transfer Matrix Method (TMM). Each layer phase shift is calculated by $\delta = 2\pi nd/\lambda$ where n is refractive indices and d is thickness of each layer and δ gives phase shift at each layer. The transfer matrix for a single layer is given by:

$$M_i = \begin{pmatrix} \cos\delta & \frac{i}{n_i}\sin\delta \\ n_i \sin\delta & \cos\delta \end{pmatrix} \quad (1)$$

The total matrix is calculated by multiplying the M_i for each layer: $M = M_1 \times M_2 \times M_3 \cdots \times M_N$. The transmission coefficient is given by $T = |1/M_{11}|^2$.

IV. Methodology

A. Structure Design and Simulation

The multilayered structure is implemented in MATLAB. The transmission spectrum is given in the range of 400–700 nm using TMM equations and the resonance wavelength is identified as its peak of the transmission.

B. Data Set Generation

A data set of 5,000 samples were generated by varying structural parameters. For these variations, the transmission spectrum was calculated using TMM equations and resonance wavelength was obtained.

C. Deep Learning Model

A deep neural network model was implemented using TensorFlow/Keras. The model is trained using the generated data set which consists of varying structural parameters as input and their resonance wavelength as output. The model was trained using supervised learning to minimize the prediction error.

V. Results and Discussion

A. Transmission Spectrum

The simulated spectrum showed the distinct peak within the photonic bandgap which confirms the presence of an edge state.

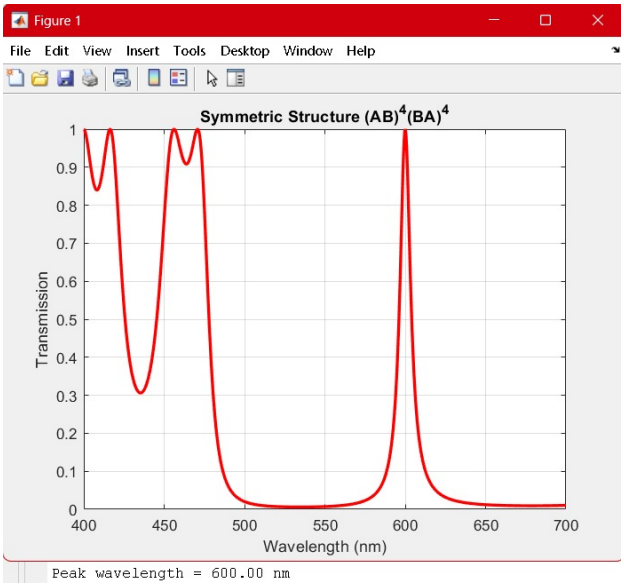


Fig. 2. MATLAB output of the transmission spectrum showing the resonance peak.

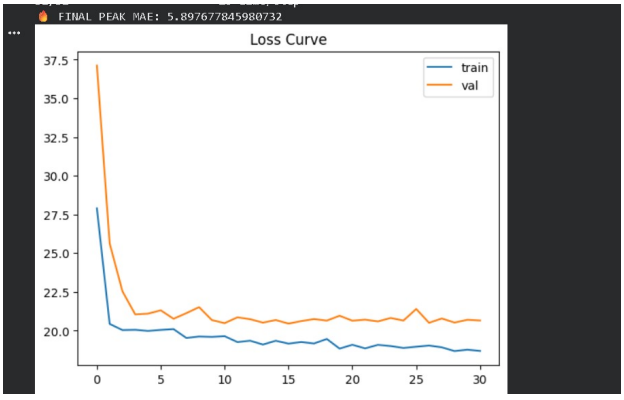


Fig. 3. Loss curve showing training and validation MAE.

B. Inverse Design Results

For a target wavelength of 620 nm, the optimized structure produced a predicted output of approximately 620.8597 nm, corresponding to an error of less than 1 nm.

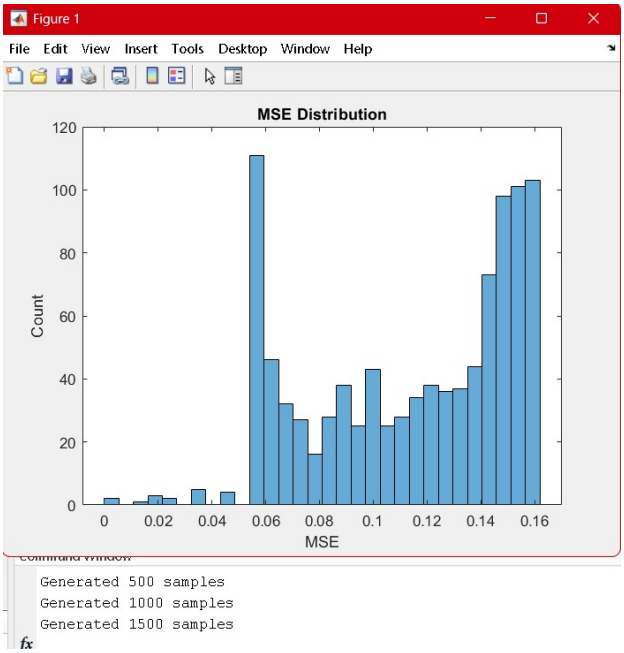


Fig. 4. MSE Distribution across the samples generated from the modified TPC code.

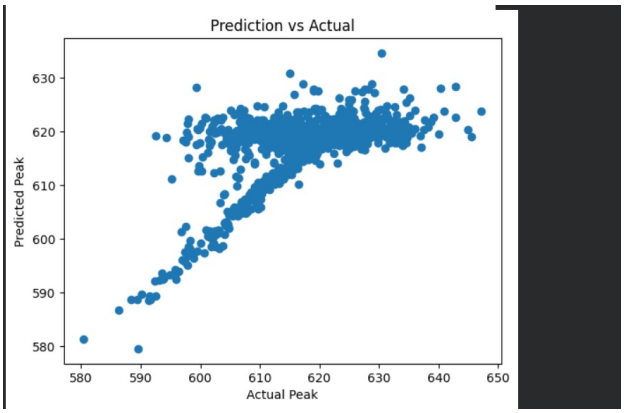


Fig. 5. Scatter plot comparing predicted resonance peaks vs. actual peaks.

VI. Conclusion

In this research, the 1D topological photonic crystal was designed and analyzed using TMM. A deep learning model (DNN) was developed to predict the resonance wavelength with reasonable accuracy, and even inverse design was successfully performed to determine structural parameters for a desired wavelength. The result combined the physics-based model with a deep learning model for efficient photonic design.

VII. Future Work

Future endeavors will focus on enhancing the deep learning model by integrating sophisticated architectures to improve prediction precision. You can also use the suggested method on two- and three-dimensional photonic crystals to get more uses in the real world. Adding constraints to the fabrication process and testing the results in real life will also help make sure they work in the real world. You can make the device work even better by optimizing multiple parameters, such as Q-factor and transmission efficiency.

References

[1] [1] T. Ozawa et al., "Topological photonics," Rev. Mod. Phys., vol. 91, no. 1, p. 015006, 2019.
[2] [2] S. Kim et al., "Robustness of the topological interface state in a 1D photonic crystal resonator with an air-gap," in Proc. CLEO/Europe-EQEC, 2021.
[3] [3] A. Singh et al., "Topological protected edge state with ultra-high Q-factor in one dimensional photonic crystal heterostructure," in IEEE WRAP, 2023.
[4] [4] G. Ali J et al., "Advancing topological robustness in 1D photonic crystals via multi-material architectures," Brazilian Journal of Physics, 2026.
[5] [5] L. Lu et al., "Topological photonics," Nature Photonics, vol. 8, no. 11, pp. 821–829, 2014.
[6] [6] Y. Ota et al., "Topological photonic crystal nanocavity laser," Communications Physics, vol. 1, no. 1, 2018.
[7] [7] M. Xiao et al., "Surface impedance and bulk band geometric phases in one-dimensional systems," Phys. Rev. X, 2014.
[8] [8] P. Yeh, Optical Waves in Layered Media, Wiley, 1988.

Deep Learning-Based Forward and Inverse Design of 1D Topological Photonic Crystal

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INTRODUCTION

- Photonic crystals are periodic dielectric structures that control the propagation of light
- They exhibit unique optical properties such as photonic bandgaps and resonance modes
- Topological photonic crystals (TPCs) support localized edge states that are robust and highly confined
- These structures are widely used in optical filtering, sensing, and communication systems

PROBLEM STATEMENT

- Designing 1D topological photonic crystal structures requires precise control of resonance wavelength
- The optical response depends on multiple parameters such as layer thickness and refractive index
- Conventional analysis using the Transfer Matrix Method (TMM) requires repeated simulations for every parameter variation
- This makes the design process computationally intensive and time-consuming

Additionally:

- There is no direct or fast method to predict the resonance wavelength from structural parameters

Therefore:

- A need arises for a fast and efficient predictive model using Deep Learning to approximate the optical behavior of the structure

OBJECTIVE

- To design a 1D topological photonic crystal (TPC) structure
- To analyze its optical response using the Transfer Matrix Method (TMM)
- To develop a deep learning model for predicting resonance wavelength
- To perform inverse design for obtaining structural parameters from a target wavelength

PHYSICAL STRUCTURE

- **Structure Used: $(AB)^4$ $(BA)^4$**
- A ⟶ Low refractive index material
- B ⟶ High refractive index material
- Periodic and symmetric multilayer arrangement

Key Insight:

- The interface between (AB) and (BA) creates a topological boundary
- This boundary supports a localized optical mode (edge state)

Result:

- Formation of a sharp resonance peak in the transmission spectrum

TRANSFER MATRIX METHOD (TMM)

- TMM is used to analyze wave propagation in multilayer optical systems
- Each layer is represented using a matrix that describes phase and amplitude changes

By multiplying matrices of all layers:

- Total system response is obtained

Output:

- Transmission spectrum as a function of wavelength

MODIFIED TPC (MTPC)

- Introduced a defect layer (dB2x) in the structure
- Allows controlled variation of structural parameters

Effect:

- Resonance wavelength shifts based on thickness variation
- Enables tunable photonic response

DATASET GENERATION

- Generated 5000 samples using MATLAB simulation

Each sample consists of:

Inputs:

- dA → Thickness of layer A
- dB → Thickness of layer B
- dB2x → Defect layer thickness

Output:

- Resonance peak wavelength
- Dataset is physics-based and simulation-driven

DEEP LEARNING MODEL

- **Model used**: Deep Neural Network (DNN)
- Learns nonlinear relationship between structure and optical response

Mapping:

- Includes multiple dense layers and nonlinear activation
- Enables fast prediction without simulation

TRAINING DETAILS

- **Platform:** Google Colab
- **Loss Function:** Huber / MSE
- **Training Epochs:** ~30–150

Features include both:

- Structural parameters
- Derived physics-based features

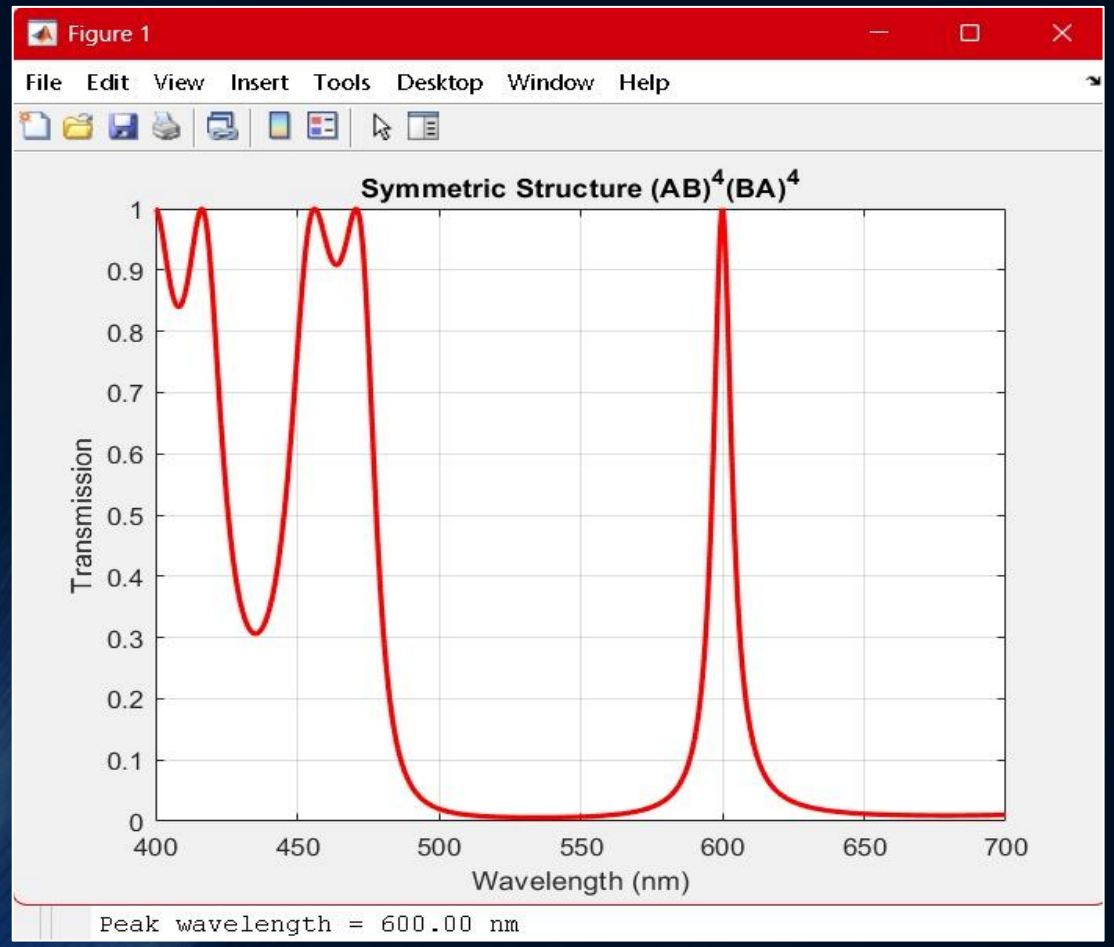
MATLAB SIMULATION

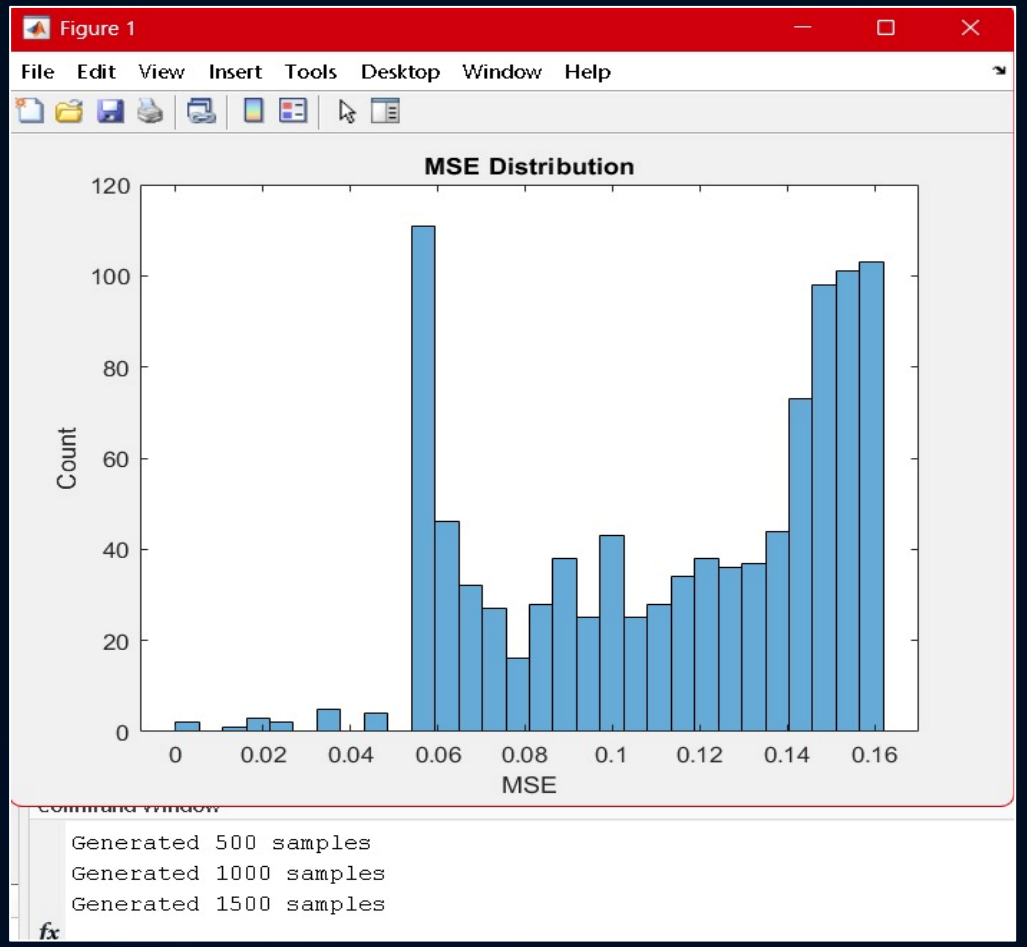
- Simulation performed over wavelength range: 400–700 nm
- Transmission spectrum calculated using TMM
- Resonance peak identified from the spectrum

Observation:

- A sharp transmission peak appears due to the topological edge state

MATLAB SIMULATION RESULTS:





RESULTS

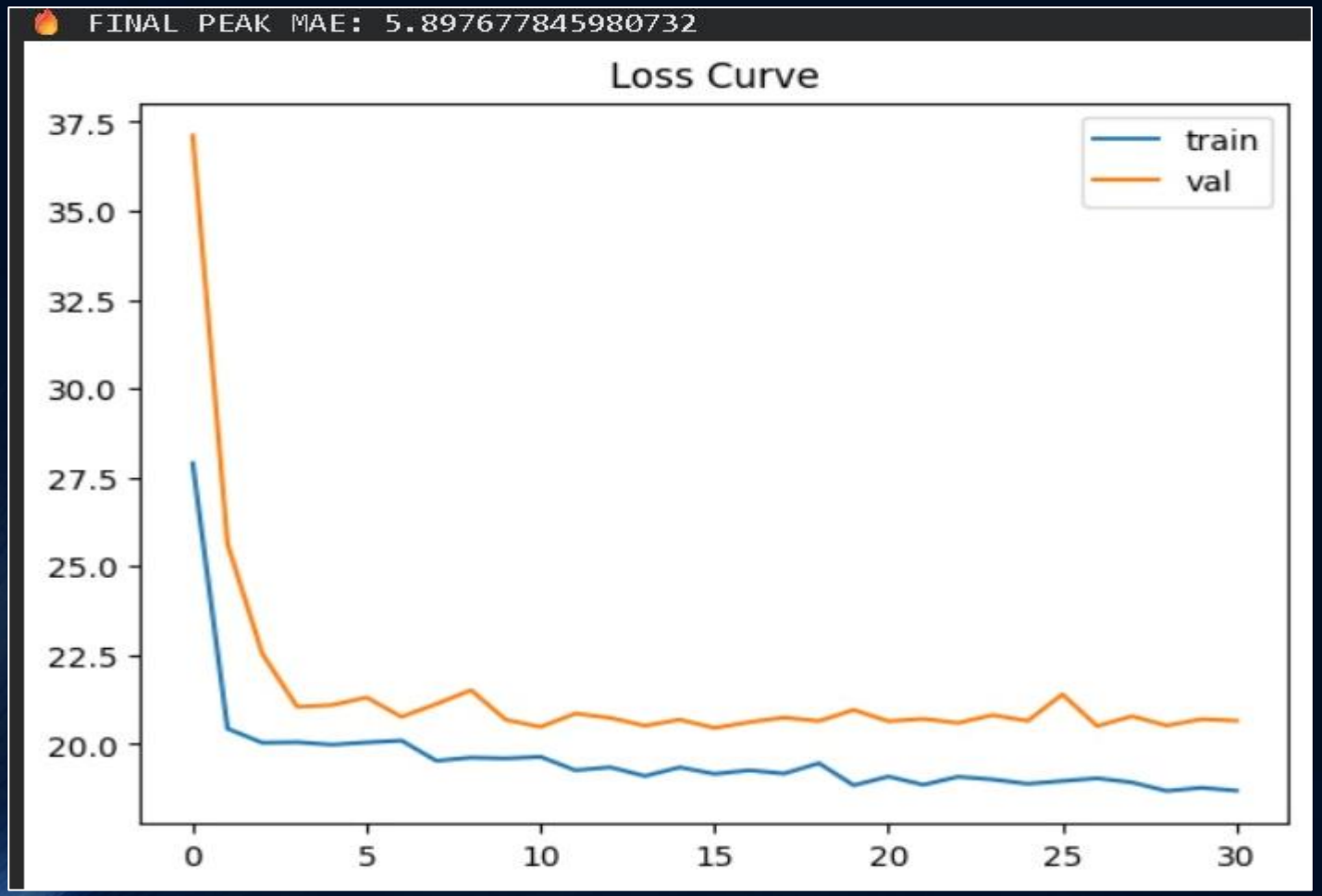
- Mean Absolute Error (MAE) ≈ 6 nm
- Root Mean Square Error (RMSE) ≈ 7–8 nm
- R^2 Score ≈ 0.3

Visualizations:

- Training Loss Curve
- Prediction vs Actual Plot

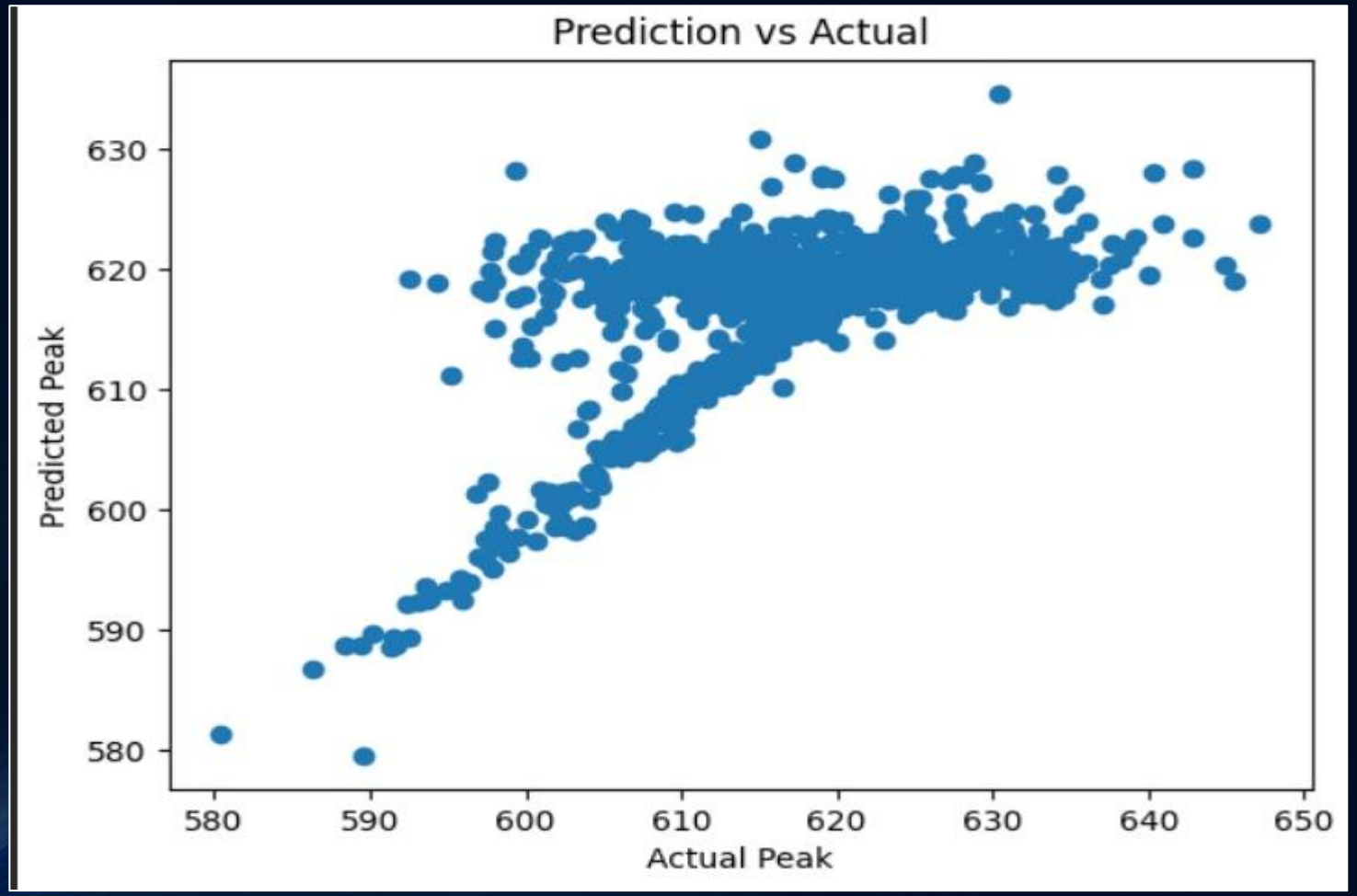
TRAINING LOSS CURVE

🔥 FINAL PEAK MAE: 5.897677845980732



📊 ADDITIONAL METRICS
MSE : 62.88141381080156
RMSE : 7.929780186789642
R2 : 0.356446748782152

PREDICTION vs ACTUAL PLOT



RESULT ANALYSIS

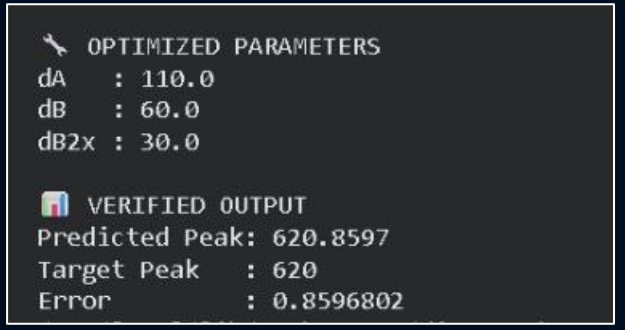
- Model successfully captures the overall trend of peak variation
- Prediction accuracy is reasonable for nonlinear physical systems

Limitations observed:

- Slight deviation at higher wavelength region

Reason:

- Complex nonlinear behavior of optical system
- Limited dataset diversity



INVERSE DESIGN

- Objective: Determine structure for a desired wavelength

Example:

- Target = 620 nm
- Predicted = 620.86 nm
- Error ≈ 0.86 nm

Significance:

- Enables design of photonic structures without repeated simulation
- Demonstrates practical application of deep learning

ADVANTAGES

- Significantly reduces computation time
- Eliminates repeated simulations
- Enables fast and efficient design
- Suitable for real-time applications

LIMITATIONS

- Prediction error around ~6 nm
- Limited dataset size
- Sensitivity of optical response to parameter variations

CONCLUSION

- Successfully designed a 1D topological photonic crystal
- Analyzed optical behavior using TMM
- Developed a deep learning model for prediction
- Achieved ~6 nm prediction accuracy
- Demonstrated effective inverse design capability

FUTURE WORK

- Extend to 2D photonic crystal structures
- Improve dataset size and diversity
- Explore advanced deep learning architectures
- Real-time photonic device implementation

Thank You!